

Medical Insurance Cost Analysis

What the Project Does

This project takes a dataset of 1,338 insurance policyholders — including their age, sex, BMI, number of children, smoking status, region, and annual medical charges — and runs it through a full statistical workflow. The goal was to explore the data, test some hypotheses, build a prediction model, check whether that model can be trusted, and finally package it all into an interactive dashboard.

What the Data Looks Like

Men and women were fairly evenly split, but there were far more non-smokers (1,064) than smokers (274). Age and BMI both followed fairly normal, symmetric distributions.

Charges, however, are heavily skewed. Most people pay modest amounts, but a small group of expensive claims drags the average way up. That's why the mean charge (\$13,270) is so much higher than the median (\$9,382) — a classic sign that 'average' numbers can be misleading here. This is also why the project was careful to test for normality before choosing which statistical tests to use.

What Actually Drives Cost

1. Smoking is the big one.

Smokers pay roughly 3.8 times more than non-smokers on average (\$32,050 vs \$8,434), and this difference is statistically massive — there's essentially no chance it's a coincidence.

2. Region matters, but with a twist.

Raw averages show the Southeast paying the most. But once smoking, BMI, and age are controlled for, the Southeast actually comes out cheaper than the Northeast. In other words, regional differences aren't really about location — they're about who happens to live there (e.g., more smokers or higher BMI in certain regions).

3. Age, BMI, and children all nudge costs up.

Each extra year of age adds about \$257 in expected charges, each BMI point adds about \$339, and each additional child adds about \$476 — all statistically meaningful effects.

4. Sex doesn't matter.

Once everything else is accounted for, being male or female showed no meaningful effect on charges.

5. Smoking and region are unrelated.

Smokers aren't more concentrated in one region than another — the two factors appear independent.

The Prediction Model

A regression model was built that explains about 75% of why charges vary from person to person — a solid result for real-world health cost data. But it isn't perfect:

- The model's errors (residuals) aren't perfectly normal and show somewhat uneven spread, a sign that a simple additive model misses some nuance — especially around how BMI and smoking seem to combine and amplify each other's effect.
- No signs of multicollinearity (the predictors aren't redundant with one another) or autocorrelation issues were found, so the model's underlying structure is sound — it just doesn't capture every curve in the data.

What Could Make It Better

- Adding a BMI $\times$ smoking interaction term, since the two factors seem to compound one another.
- Trying a log-transformed model or a Gamma-distribution model, both better suited to skewed cost data.
- Including missing factors like pre-existing conditions, family history, or lifestyle habits.

- Testing whether machine learning models (Random Forest, XGBoost) outperform the linear approach.

The Dashboard

An interactive Streamlit app was also built, allowing someone to filter the data, run the hypothesis tests live, and plug in a person's profile (age, BMI, smoker status, etc.) to get a predicted charge — along with a confidence range for the 'average' person like them, and a wider range for what an individual might actually pay.